

EBN111 EKSAMEN/EXAMINATION (17 JUNIE/JUNE 2014)

ANTWOORDBLAD / ANSWER SHEET

Van / Surname:	Memo	Student #:							
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Q1		Q2		Q3		Q4		Tot:	/90
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VRAAG 1 – QUESTION 1 (20)

1.1	a) $R_{savings} = 35040$	b) $W_{Total} = 21.5$	c) $V_{battery} = 3.5V$
1.2	$P_1 = 180W$	$P_2 = 240W$	$v_1 = -50V$ $i_2 = 8A$
1.3	$v_{xy} = 9V$		$i_x = 1A$
1.4	$R = 5\Omega$		$i_0 = 4A$
1.5	$R_{ab} \text{ (Circuit 1)} = 50\Omega$		$R_{ab} \text{ (Circuit 2)} = 21\Omega$
1.6	$R_{cd} = 30\Omega$		

VRAAG 2 – QUESTION 2 (33)

2.1	a = 3	b = 0	c = 10	d = -25
2.2	e = 300	f = 385	g = 1	h = -1
2.3	$\overset{m}{k} = 12$		$V = -0.25$	
2.4	$V_0 = 19.8V$			
2.5	$\begin{bmatrix} 0.2 & -0.2 & 0 \\ -0.2 & 0.4 & -0.1 \\ 0 & -0.1 & 0.1 \end{bmatrix} \cdot \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 6 \\ 0 \\ 15 \end{bmatrix}$			
2.6	a) $I_E = 71.6mA$		b) $I_B = 3.36\mu A$	
2.7	a) $v_0 (i_s=2A) = 4V$		b) $v_0 (i_s=8mA) = 16mV$	
2.8	$v_s = 2.22V$			
2.9	$v_y = -3.33V$			
2.10	$V_{TH} = 80.5V$		$R_{TH} = 40\Omega$	
2.11	$I_N = 5.33A$		$R_N = 75\Omega$	

VRAAG 3 – QUESTION 3 (17)			
3.1	$v_o = 4V$	$i_o = -1mA$	
3.2	a) $R_f = 5.14k\Omega$	b) $i_o = 1mA$	b) $v_o = 8V$
3.3	a) $v_o = 4mV$	b) $R_1 = 2k\Omega$	b) $R_2 = 4k\Omega$
3.4	$v_1 = -20mV$	$v_2 = -2mV$	$v_o = -36mV$
3.5	$w_L = 20\mu J$	$w_C = 72\mu J$	
3.6	a) $v_C = 333.3mV$	b) $v_{L=5H} = 99.5\mu V$	

VRAAG 4 – QUESTION 4 (20)			
4.1	a) $v_o(t) = 10\cos(10t + 90^\circ) V$		
	b) $\underline{v_1}$ leads $\underline{v_2}$ by 120°		
4.2	$\bar{I}_1 = 26 \angle 15^\circ$ (polar)	$\bar{V}_1 = 1 \angle 165^\circ$ (polar)	
4.3	$i_X(t) = 1.768\cos(10t + 45^\circ) A$		
4.4	$Z_{in} = 15 - 16j \Omega$	(rectangular)	
4.5	$\bar{I}_Y = -0.32 + 0.24j$	(rectangular)	
4.6	$\bar{V}_0 = 12 - 24j V$	(rectangular)	
4.7	$\bar{V}_{TH} = 4 \angle -135^\circ V$ (polar)	$Z_{TH} = 2 + 2j \Omega$	(rectangular)